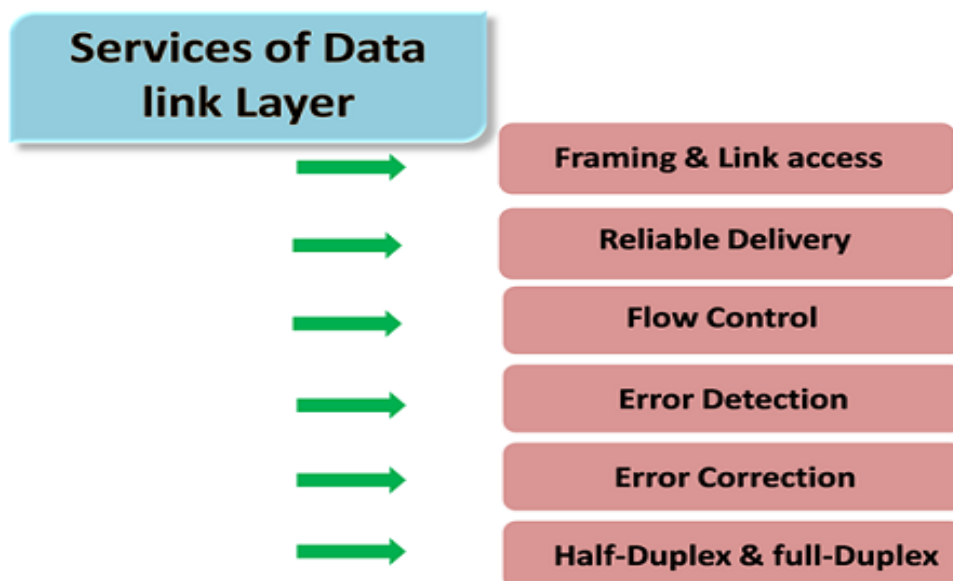


MODULE-4

Introduction to link layer - Error detection (parity, checksum, and CRC). Multiple access protocols (collision and token based). IEEE 802.3 Ethernet, Switching and bridging, Data encoding. Ethernet switches.

INTRODUCTION TO DATA-LINK LAYER

- In the OSI Model, the Data Link Layer is the **2nd layer from the bottom** and **4th from the top**.
- The connection between two nearby devices is called a **link**. To send data from source to destination, the data (datagram) must pass through each link step by step.
- The **main job of the Data Link Layer** is to move the datagram across a **single link** between two directly connected devices.
- The Data Link Layer protocol decides:
- How the data packet should look (format)
- What actions to take, such as:
 - Error detection
 - Retransmission
 - Flow control
 - Random access
- Some common Data Link Layer protocols are:
- Ethernet
- Token Ring
- FDDI (Fibre Distributed Data Interface)
- PPP(point-to-point)



- **Framing & Link Access:**

The Data Link Layer takes data from the Network Layer and wraps it into a frame before sending it.

A frame contains:

- The actual data (datagram)
- Some extra information (fields)

It also decides how and when devices can send data on the link.

- **Reliable Delivery:**

The Data Link Layer tries to send data without errors.

It uses:

- Acknowledgements (confirmation from receiver)
- Retransmission (send again if error occurs)
this is especially useful because errors can be fixed at the same link, instead of sending data again from the beginning.

- **Flow Control:**

Sometimes the receiver is slower than the sender.

Without control, data may be lost due to buffer overflow.

So, the Data Link Layer controls the speed of sending to match the receiver.

- **Error Detection:**

Errors can happen due to noise or signal loss.

The Data Link Layer adds extra bits to check if data is correct.

The receiver uses these bits to detect errors.

- **Error Correction:**

In this case, the receiver not only finds the error, but also corrects it without asking for retransmission.

- **Half-Duplex & Full-Duplex:**

- Full-Duplex: Both devices can send data at the same time
- Half-Duplex: Only one device can send at a time

Error Detection

When data is transmitted from one device to another device, the system does not guarantee whether the data received by the device is identical to the data transmitted by another device. An Error is a situation when the message received at the receiver end is not identical to the message transmitted.

The duration of noise in Burst Error is more than the duration of noise in Single-Bit. Burst Errors are most likely to occur in Serial Data Transmission. The number of affected bits depends on the duration of the noise and data rate.

Error Detecting Techniques:

- Single parity check
- Two-dimensional parity check
- Checksum
- Cyclic redundancy check

Single parity check

A Single Parity Check is one of the simplest and earliest techniques used for error detection in data communication systems. It is based on the principle of adding one extra bit, called a *parity bit*, to a block of binary data.

This technique belongs to the domain of Data Communication and is widely discussed in error control coding mechanisms.

The main objective is to ensure that the transmitted data maintains a specific parity condition:

- Even parity → total number of 1s (including parity bit) is even
- Odd parity → total number of 1s is odd

Working Principle

The sender calculates the parity bit based on the number of 1s in the data unit.

- If using even parity:
 - If number of 1s is already even → parity bit = 0
 - If number of 1s is odd → parity bit = 1
- If using odd parity:
 - If number of 1s is already odd → parity bit = 0
 - If number of 1s is even → parity bit = 1

The parity bit is appended to the data before transmission.

Example (Even Parity)

Consider the data:

Data = 1011001

Step 1: Count number of 1s

Number of 1s = 4 (even)

Step 2: Determine parity bit (Even parity)

Since 4 is already even $\rightarrow$ parity bit = 0

Step 3: Transmitted data:

1011001 0

Receiver Side Operation

At the receiver:

- Count the number of 1s in the received data (including parity bit)
- Check whether parity condition is satisfied

Case 1: No Error

Received = 10110010

Number of 1s = 4 $\rightarrow$ Even $\rightarrow$ No error

Case 2: Error Occurs

suppose one bit is changed during transmission:

Received = 10111010

Number of 1s = 5 $\rightarrow$ Odd $\rightarrow$ Error detected

Detection Capability

Single parity check can:

- Detect single-bit errors
- Cannot detect even number of bit errors (e.g., 2-bit errors)

Example of failure:

Original: 10110010

Error: 10010010 (two bits flipped)

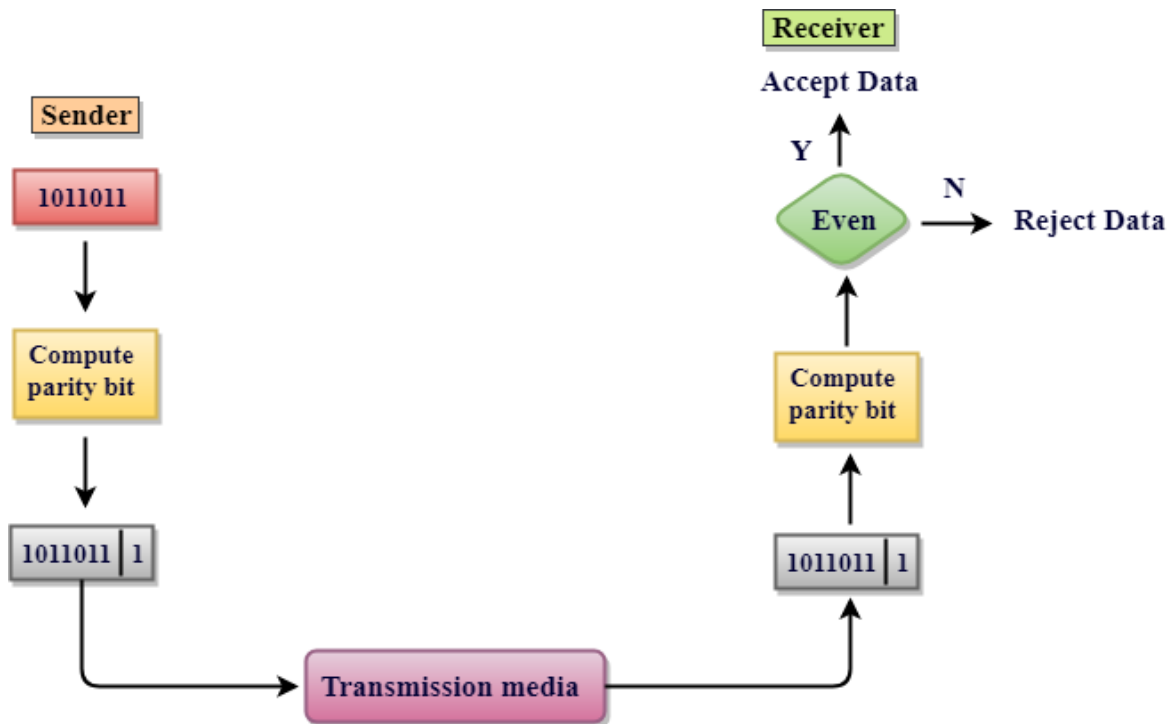
Number of 1s still even $\rightarrow$ Error NOT detected

Advantages

- Very simple and easy to implement
- Low overhead (only 1 extra bit)
- Fast error detection mechanism

Limitations

- Cannot correct errors (only detects)
- Fails for multiple-bit errors (especially even-bit errors)
- Not reliable for high-noise communication systems



Two-Dimensional Parity Check

1. Concept Overview

A Two-Dimensional Parity Check is an advanced extension of the single parity method used in Data Communication. It improves error detection capability by organizing data into a matrix (rows and columns) and applying parity checks in two directions:

- Row-wise parity (Horizontal check) → detects errors in each row
- Column-wise parity (Vertical check) → detects errors in each column

This dual checking significantly increases reliability compared to single parity.

2. Working Principle

The sender arranges the data bits into a table (matrix), then:

1. Adds a parity bit to each row
2. Adds a parity bit to each column

3. Sometimes, an additional overall parity bit is added

Thus, redundancy is introduced in both dimensions.

3. Example (Even Parity)

Let us take 4 data units (each of 7 bits):

Data:

1011001

1100101

1001110

0110110

Step 1: Arrange in Matrix

1 0 1 1 0 0 1

1 1 0 0 1 0 1

1 0 0 1 1 1 0

0 1 1 0 1 1 0

Step 2: Add Row Parity (Even Parity)

Count 1s in each row:

- Row 1 $\rightarrow$ 4 ones $\rightarrow$ parity = 0
- Row 2 $\rightarrow$ 4 ones $\rightarrow$ parity = 0
- Row 3 $\rightarrow$ 4 ones $\rightarrow$ parity = 0
- Row 4 $\rightarrow$ 4 ones $\rightarrow$ parity = 0

Updated matrix:

1 0 1 1 0 0 1 | 0

1 1 0 0 1 0 1 | 0

1 0 0 1 1 1 0 | 0

0 1 1 0 1 1 0 | 0

Step 3: Add Column Parity

Now calculate parity for each column:

Column parity row:

$(1+1+1+0)=3 \rightarrow$ parity=1

$(0+1+0+1)=2 \rightarrow$ parity=0

$(1+0+0+1)=2 \rightarrow$ parity=0

$(1+0+1+0)=2 \rightarrow$ parity=0

$(0+1+1+1)=3 \rightarrow \text{parity}=1$
 $(0+0+1+1)=2 \rightarrow \text{parity}=0$
 $(1+1+0+0)=2 \rightarrow \text{parity}=0$
 (Row parity column also checked)

Final transmitted block:

```

1 0 1 1 0 0 1 | 0
1 1 0 0 1 0 1 | 0
1 0 0 1 1 1 0 | 0
0 1 1 0 1 1 0 | 0
-----
1 0 0 0 1 0 0 | 0 ← column parity row
  
```

4. Error Detection at Receiver

The receiver:

- Recalculates row parity
- Recalculates column parity

Case: Single Bit Error Detection

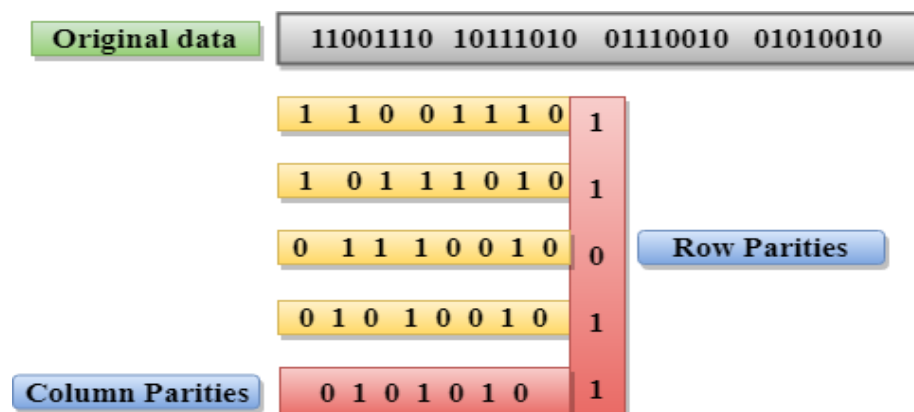
Suppose a bit changes:

Original: 1 → Error: 0

Example error at row 2, column 3:

- Row 2 parity becomes incorrect
- Column 3 parity becomes incorrect

The intersection of faulty row and column pinpoints the exact error location



6. Advantages

- More reliable than single parity
- Can **locate the exact position of a single-bit error**
- Simple implementation compared to advanced techniques

7. Limitations

- Still not perfect for complex burst errors
- Overhead increases (extra bits for rows and columns)
- Less efficient than modern methods like CRC

Two-dimensional parity represents an important transitional technique between simple error detection and sophisticated correction methods. While it introduces spatial redundancy (matrix structure), its inability to consistently detect all multiple-bit errors limits its application in modern high-speed networks.

Checksum

The **Checksum** is a widely used error detection method in **Data Communication**. Unlike simple parity, checksum works on **groups of bits (blocks of data)**, making it more reliable for detecting transmission errors.

In real-world communication systems, data is not sent bit-by-bit but in **blocks (frames or segments)**. During transmission, noise can alter bits. The checksum method introduces **redundancy** by computing a value based on the entire data block.

The core idea:

- Sender calculates a value (checksum) from data
- Sends **data + checksum**
- Receiver recalculates and verifies

If both results match → data is assumed correct.

2. One's Complement Arithmetic (Key Idea)

Checksum uses **one's complement addition**, which is slightly different from normal binary addition:

- Add binary numbers normally
- If there is a **carry from the most significant bit**, add it back to the result

This is called **end-around carry**

3. Checksum Generation (Sender Side – Detailed Steps)

Let us understand this carefully:

Step 1: Divide Data

The data block is divided into **k segments**, each of **n bits**.

Example:

Data = 100110011001

Divide into 4-bit parts:

1001 1001 1001

Step 2: Add All Segments (One's Complement Addition)

```
1001
+ 1001
+ 1001
-----
sum
```

Perform binary addition:

- If overflow occurs → add carry back to result

Step 3: Complement the Sum

Convert all bits:

- 1 → 0
- 0 → 1

This result is the **checksum**

Step 4: Send Data

Final transmitted message:

Original Data + Checksum

4. Checksum Verification (Receiver Side – Detailed Steps)

When the receiver gets the data:

Step 1: Divide Again

Split received data (including checksum) into same segments

Step 2: Add All Segments

Use **one's complement addition** again

Step 3: Complement the Result

Take complement of final sum

Step 4: Decision Rule

Result after complement	Meaning
All 0s	No error
Not all 0s	Error present

5. Complete Worked Example

Sender Side

Assume 4-bit segments:

Data:

1010

1100

0110

Step 1: Add

1010

+ 1100

= 10110(carry generated)

Add carry:

0110 + 1 = 0111

Now add third segment:

0111

+ 0110

= 1101

Step 2: Complement

1101 → 0010

Checksum = 0010

Step 3: Transmit

Data + Checksum:

1010 1100 0110 0010

Receiver Side

Step 1: Add All

1010

1100

0110

0010

Add step-by-step → result becomes:

1111

Step 2: Complement

1111 → 0000

Result is all zeros → **No error**

6. Error Case Example

If one bit changes during transmission:

Received:

1010

1101(error here)

0110

0010

After addition and complement:

Result ≠ 0000

Error detected

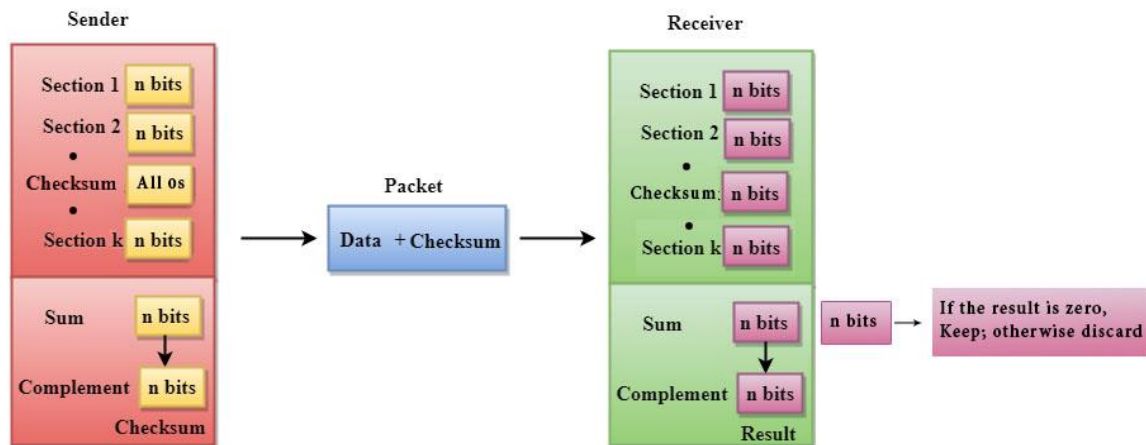
7. Advantages

- More powerful than parity check
- Works on blocks (better detection)

- Simple implementation in hardware and software
- Used in protocols like **IP, TCP, UDP**

8. Limitations

- Cannot detect all types of errors (especially some multiple-bit errors)
- Less reliable than **CRC (Cyclic Redundancy Check)**
- No error correction (only detection)

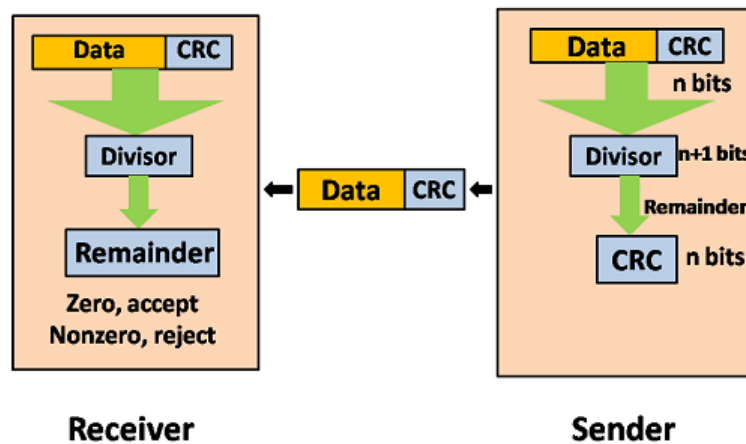


Cyclic Redundancy Check(CRC)

1. Add zeros to data
First, some zeros are added to the end of the data.
The number of zeros added is one less than the number of bits in the divisor.
2. Divide the data
Now, this new data (data + zeros) is divided by a fixed binary number called the divisor using binary division (XOR method).
3. Find the remainder
the result left after division is called the CRC remainder.
4. Attach remainder to data
Replace the added zeros with this remainder.
Now, the final data (data + remainder) is sent to the receiver.
5. Receiver checks the data
the receiver takes the full received data and divides it again using the same divisor.

6. Final result

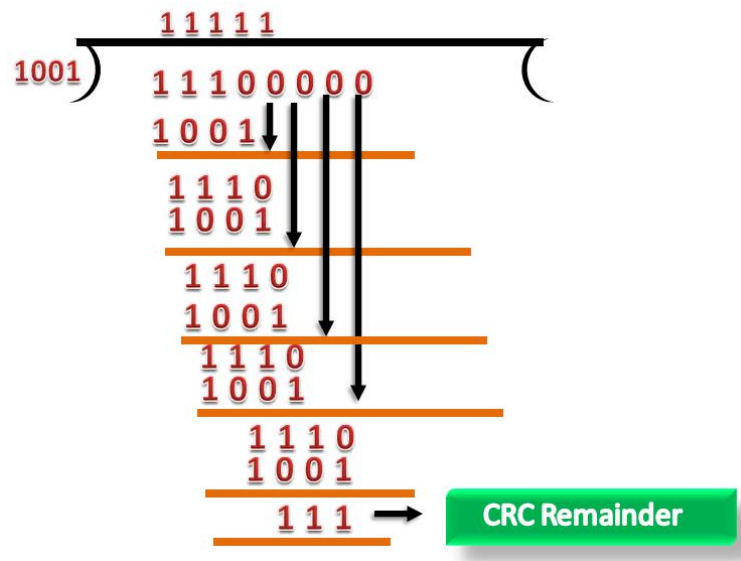
- If remainder = 0 → No error, data is correct
- If remainder $\neq$ 0 → Error, data is rejected



Suppose the original data is 11100 and divisor is 1001.

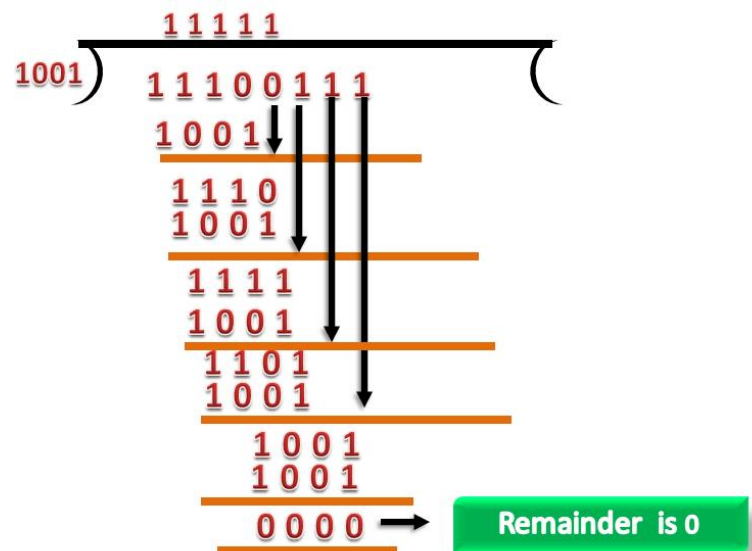
CRC Generator

- A CRC generator uses a modulo-2 division. Firstly, three zeroes are appended at the end of the data as the length of the divisor is 4 and we know that the length of the string 0s to be appended is always one less than the length of the divisor.
- Now, the string becomes 11100000, and the resultant string is divided by the divisor 1001.
- The remainder generated from the binary division is known as CRC remainder. The generated value of the CRC remainder is 111.
- CRC remainder replaces the appended string of 0s at the end of the data unit, and the final string would be 11100111 which is sent across the network.



CRC Checker

- The functionality of the CRC checker is similar to the CRC generator.
- When the string 11100111 is received at the receiving end, then CRC checker performs the modulo-2 division.
- A string is divided by the same divisor, i.e., 1001.
- In this case, CRC checker generates the remainder of zero. Therefore, the data is accepted.



Multiple Access Protocols-Collision and Token Based

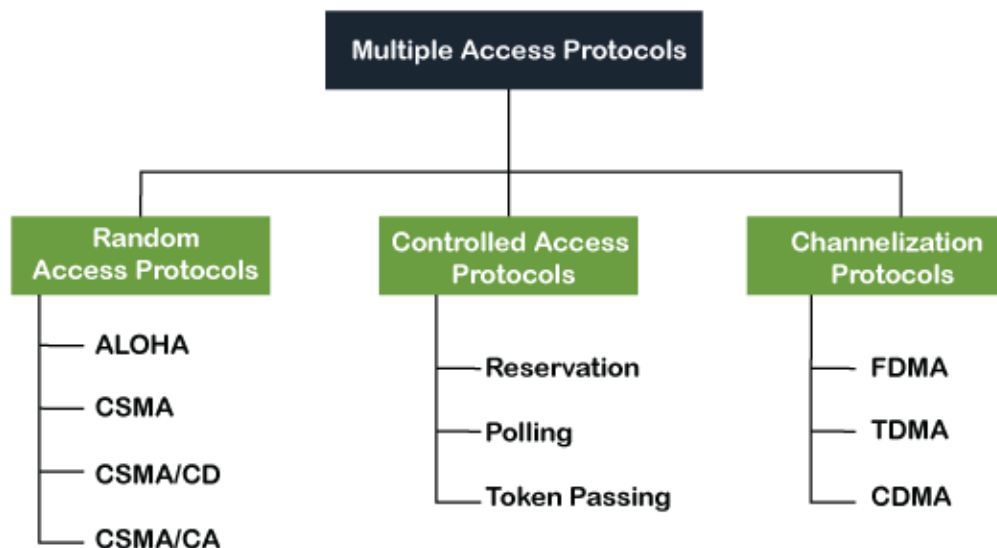
The data link layer is used in a computer network to transmit the data between two devices or nodes. It divides the layer into parts such as **data link control** and the **multiple access resolution/protocol**. The upper layer has the responsibility to flow control and the error control in the data link layer, and hence it is termed as **logical of data link control**. Whereas, the lower sub-layer is used to handle and reduce the collision or multiple access on a channel. Hence it is termed as media access control or the multiple access resolutions.

Data Link Control

A data link control is a reliable channel for transmitting data over a dedicated link using various techniques such as framing, error control and flow control of data packets in the computer network.

What is a multiple access protocol?

When a sender and receiver have a dedicated link to transmit data packets, the data link control is enough to handle the channel. Suppose there is no dedicated path to communicate or transfer the data between two devices. In that case, multiple stations access the channel and simultaneously transmit the data over the channel. It may create collision and cross talk. Hence, the multiple access protocol is required to reduce the collision and avoid crosstalk between the channels.



Random Access Protocol

Random Access Protocol is a **medium access control technique** in which **all stations have equal rights** to access the communication channel. There is **no central controller**, and each station independently decides when to transmit data.

In this method, a station first observes the state of the channel (idle or busy) and then attempts transmission. However, since multiple stations operate independently, **simultaneous transmission may occur**, leading to **collisions**. These collisions can result in **data loss, corruption, or retransmission**, thereby reducing network efficiency.

Key Characteristics

- Equal priority for all stations
- No centralized control
- High probability of collisions
- Requires retransmission mechanisms

Types of Random Access Protocols

- ALOHA
- CSMA
- CSMA/CD
- CSMA/CA

1. ALOHA Protocol

ALOHA is one of the **earliest random access protocols**, originally developed for **wireless communication systems**. It allows stations to transmit data **whenever they have data to send**, without checking the channel status.

Working Principle

A station sends data immediately. If the transmission is successful, an acknowledgment (ACK) is received. If not, the station assumes a collision occurred and retransmits after a **random time delay**.

Important Rules

1. Transmission can occur at any time without restriction.
2. No **carrier sensing** (channel checking) is performed.

3. Collisions are common due to simultaneous transmissions.
4. Acknowledgment is used to confirm successful delivery.
5. Retransmission is required after collision with random delay.

Limitations

- High collision rate
- Low efficiency
- Suitable mainly for low-traffic networks

2. CSMA (Carrier Sense Multiple Access)

CSMA improves upon ALOHA by introducing **carrier sensing**, meaning a station listens to the channel before transmitting.

Working Principle

- If the channel is **idle** → **transmit immediately**
- If the channel is **busy** → **wait**

This significantly reduces collisions compared to ALOHA.

CSMA Access Modes

It is a **carrier sense multiple access** based on media access protocol to sense the traffic on a channel (idle or busy) before transmitting the data. It means that if the channel is idle, the station can send data to the channel. Otherwise, it must wait until the channel becomes idle. Hence, it reduces the chances of a collision on a transmission medium.

1. 1-Persistent CSMA

- The station continuously monitors the channel.
- As soon as the channel becomes idle, it **immediately transmits**.

Advantage: Fast transmission

Disadvantage: Higher collision probability when multiple stations are waiting

2. Non-Persistent CSMA

- If the channel is busy, the station waits for a **random time** before checking again.

Advantage: Reduces collisions

Disadvantage: Increased delay

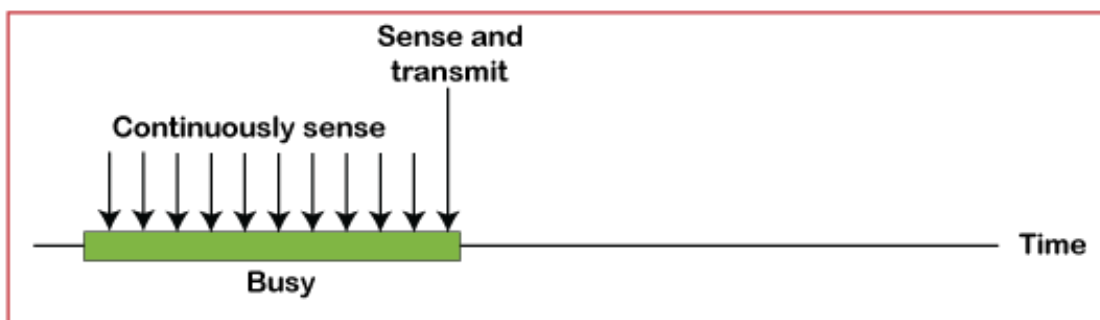
3. P-Persistent CSMA

- Used in **slotted channels**.
- When the channel is idle:
 - Transmit with probability p
 - Wait with probability $(1 - p)$

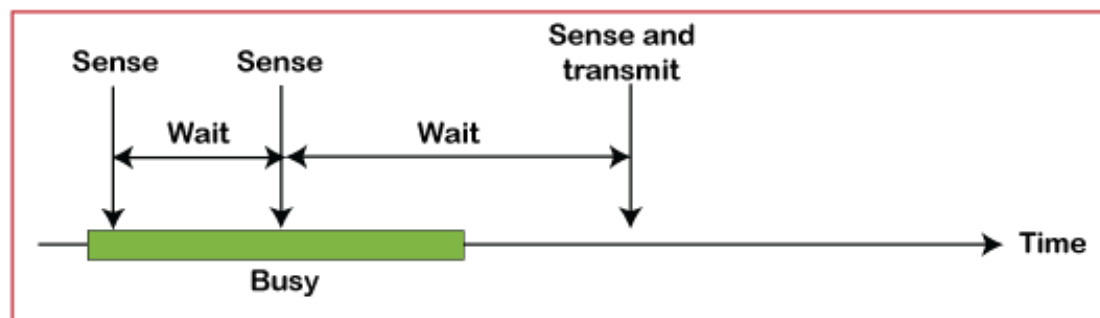
Advantage: Balances delay and collision probability

4. O-Persistent CSMA

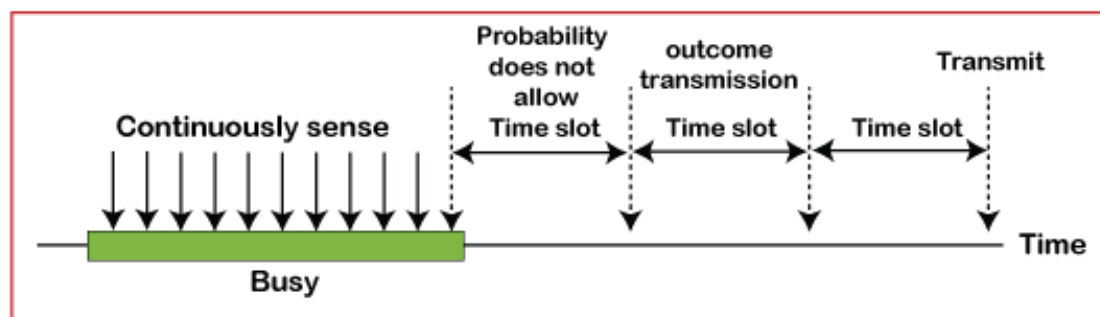
- Transmission is based on **priority order**.
- Each station waits for its assigned turn.



a. 1-persistent



b. Nonpersistent



c. p-persistent

3. CSMA/CD (Collision Detection)

It is a **carrier senses multiple access/ collision detection** network protocol to transmit data frames. The CSMA/CD protocol works with a medium access control layer. Therefore, it first senses the shared channel before broadcasting the frames, and if the channel is idle, it transmits a frame to check whether the transmission was successful. If the frame is successfully received, the station sends another frame. If any collision is detected in the CSMA/CD, the station sends a jam/ stop signal to the shared channel to terminate data transmission. After that, it waits for a random time before sending a frame to a channel.

Working Mechanism

1. Sense the channel.
2. If idle → transmit data.
3. While transmitting, monitor the channel for collisions.
4. If collision detected:
 - Send a **jam signal** to stop all transmissions
 - Wait for a **random back off time**
 - Retransmit

Key Feature

- Detects collisions **after they occur**

Limitation

- Not suitable for wireless networks (cannot detect collisions easily)

4. CSMA/CA (Collision Avoidance)

CSMA/CA is designed for **wireless networks (e.g., Wi-Fi)** where collision detection is difficult.

It is a **carrier sense multiple access/collision avoidance** network protocol for carrier transmission of data frames. It is a protocol that works with a medium access control layer. When a data frame is sent to a channel, it receives an acknowledgment to check whether the channel is clear.

If the station receives only a single (own) acknowledgments, that means the data frame has been successfully transmitted to the receiver. But if it gets two signals (its own and one more in which the collision of frames), a collision of

the frame occurs in the shared channel. Detects the collision of the frame when, a sender receives an acknowledgment signal.

Working Principle

Instead of detecting collisions, it **prevents them from occurring**.

Collision Avoidance Techniques

1. Interframe Space (IFS)

- After the channel becomes idle, a station waits for a fixed time.
- Helps prioritize transmissions.

2. Contention Window

- Time is divided into slots.
- Station selects a **random slot** before transmitting.

3. Acknowledgment (ACK)

- Receiver sends acknowledgment after successful reception.
- If no ACK is received → retransmission occurs.

Controlled Access Protocol

Controlled Access Protocol ensures that **only one station transmits at a time**, thereby eliminating collisions.

- Stations **coordinate and agree** before transmission.
- A station can transmit only when **granted permission**.
- It is a method of reducing data frame collision on a shared channel. In the controlled access method, each station interacts and decides to send a data frame by a particular station approved by all other stations. It means that a single station cannot send the data frames unless all other stations are not approved. It has three types of controlled access: **Reservation**, **Polling**, and **Token Passing**.

Types

1. Reservation

- Stations reserve time slots in advance.

2. Polling

- A central controller invites each station to transmit.

3. Token Passing

- A special token circulates among stations.
- Only the station holding the token can transmit.

Channelization Protocols

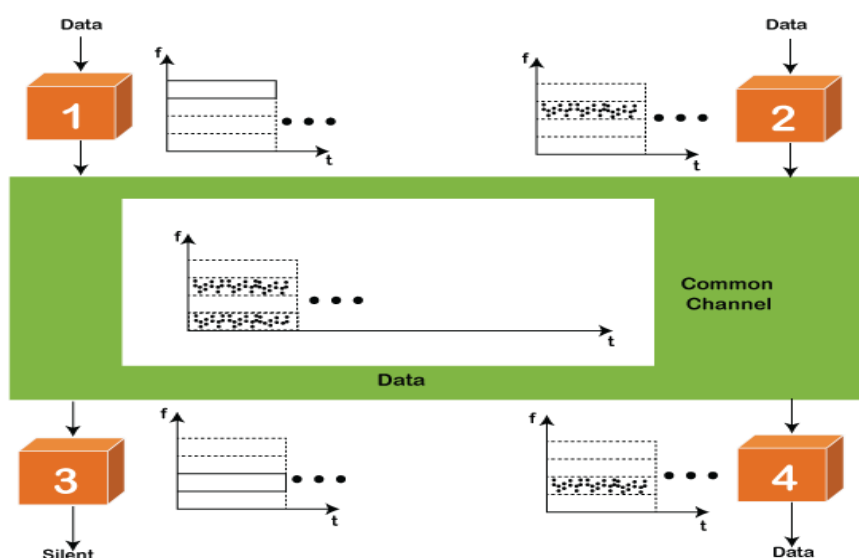
Channelization divides the communication channel into **multiple logical sub channels**, allowing simultaneous transmission without interference.

Types

- FDMA
- TDMA
- CDMA

FDMA

It is a frequency division multiple access (**FDMA**) method used to divide the available bandwidth into equal bands so that multiple users can send data through a different frequency to the sub channel. Each station is reserved with a particular band to prevent the crosstalk between the channels and interferences of stations.



TDMA

Time Division Multiple Access (**TDMA**) is a channel access method. It allows the same frequency bandwidth to be shared across multiple stations. And to avoid collisions in the shared channel, it divides the channel into different frequency slots that allocate stations to transmit the data frames. The same **frequency** bandwidth into the shared channel by dividing the signal into various time slots to transmit it. However, TDMA has an overhead of synchronization that specifies each station's time slot by adding synchronization bits to each slot.

CDMA

The Code Division Multiple Access is a channel access method. In CDMA, all stations can simultaneously send the data over the same channel. It means that it allows each station to transmit the data frames with full frequency on the shared channel at all times. It does not require the division of bandwidth on a shared channel based on time slots. If multiple stations send data to a channel simultaneously, their data frames are separated by a unique code sequence. Each station has a different unique code for transmitting the data over a shared channel. For example, there are multiple users in a room that are continuously speaking. Data is received by the users if only two-person interact with each other using the same language. Similarly, in the network, if different stations communicate with each other simultaneously with different code language.

IEEE 802.3 and Ethernet

Ethernet is a set of technologies and protocols that are used primarily in LANs. It was first standardized in 1980s by IEEE 802.3 standard. IEEE 802.3 defines the physical layer and the medium access control (MAC) sub-layer of the data link layer for wired Ethernet networks.

Ethernet is classified into two categories:

- classic Ethernet
- Switched Ethernet

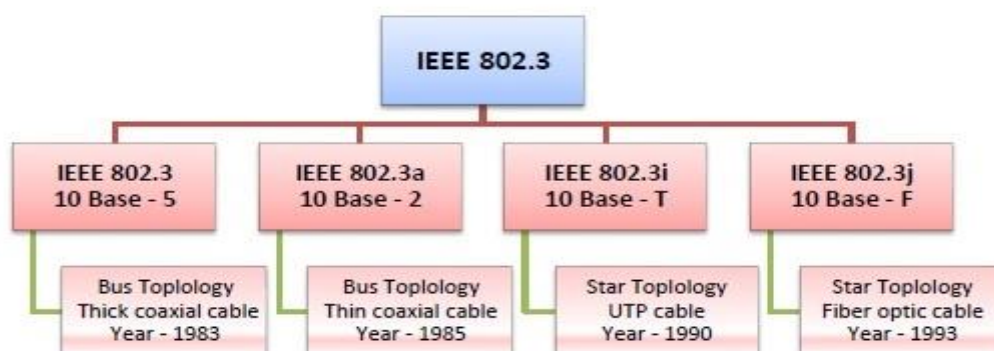
Classic Ethernet is the original form of Ethernet that provides data rates between 3 to 10 Mbps. The varieties are commonly referred as 10BASE-X. Here, 10 is the maximum throughput, i.e. 10 Mbps, BASE denoted use of baseband transmission, and X is the type of medium used. Most varieties of classic Ethernet have become obsolete in present communication scenario.

A **switched Ethernet** uses switches to connect to the stations in the LAN. It replaces the repeaters used in classic Ethernet and allows full bandwidth utilization.

IEEE 802.3 Popular Versions

There are a number of versions of IEEE 802.3 protocol. The most popular ones are -

- **IEEE 802.3:** This was the original standard given for 10BASE-5. It used a thick single coaxial cable into which a connection can be tapped by drilling into the cable to the core. Here, 10 is the maximum throughput, i.e. 10 Mbps, BASE denoted use of baseband transmission, and 5 refers to the maximum segment length of 500m.
- **IEEE 802.3a:** This gave the standard for thin coax (10BASE-2), which is a thinner variety where the segments of coaxial cables are connected by BNC connectors. The 2 refers to the maximum segment length of about 200m (185m to be precise).
- **IEEE 802.3i:** This gave the standard for twisted pair (10BASE-T) that uses unshielded twisted pair (UTP) copper wires as physical layer medium. The further variations were given by IEEE 802.3u for 100BASE-TX, 100BASE-T4 and 100BASE-FX.
- **IEEE 802.3j:** This gave the standard for Ethernet over Fibre (10BASE-F) that uses fibre optic cables as medium of transmission.

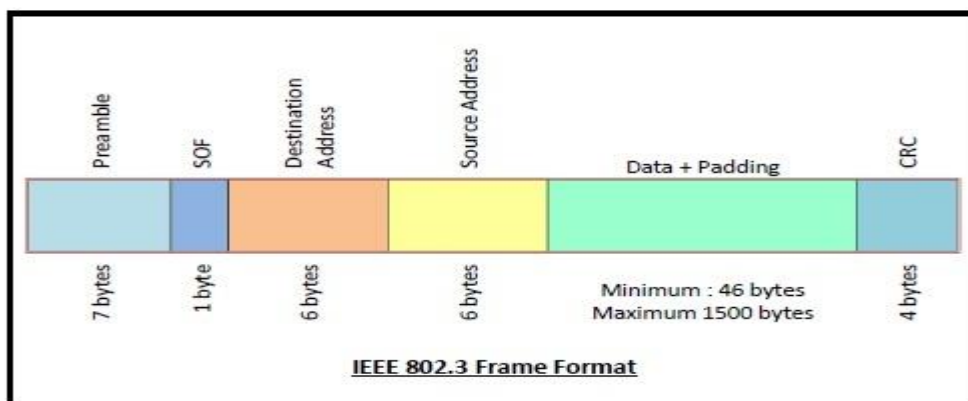
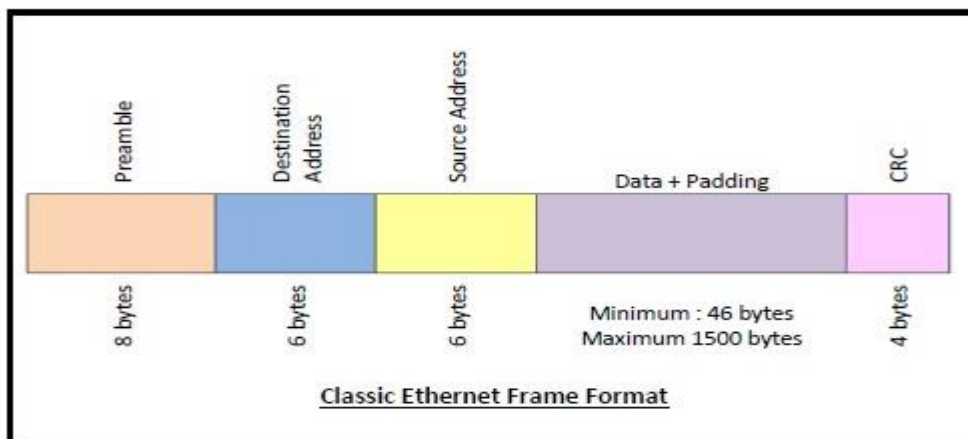


Frame Format of Classic Ethernet and IEEE 802.3

The main fields of a frame of classic Ethernet are –

- **Preamble:** It is the starting field that provides alert and timing pulse for transmission. In case of classic Ethernet, it is an 8-byte field and in case of IEEE 802.3 it is of 7 bytes.

- **Start of Frame Delimiter:** It is a 1-byte field in an IEEE 802.3 frame that contains an alternating pattern of ones and zeros ending with two ones.
- **Destination Address:** It is a 6-byte field containing physical address of destination stations.
- **Source Address:** It is a 6-byte field containing the physical address of the sending station.
- **Length:** It a 7 bytes field that stores the number of bytes in the data field.
- **Data:** This is a variable sized field carries the data from the upper layers. The maximum size of data field is 1500 bytes.
- **Padding:** This is added to the data to bring its length to the minimum requirement of 46 bytes.
- **CRC:** CRC stands for cyclic redundancy check. It contains the error detection information.



Switching and Bridging

In a computer network, many devices (computers, laptops, etc.) are connected together. If every device sends data to all others, the network becomes slow and congested.

To solve this problem, devices like bridges and switches are used. They work at the Data Link Layer (Layer 2) of the OSI Model and help in sending data only to the correct destination.

Bridging

Computer networks form the backbone of modern information systems, enabling communication, resource sharing, and distributed computing. As networks expanded, issues such as network congestion, collisions, and inefficient bandwidth usage became prominent. To address these challenges, devices like bridges were introduced.

A bridge is a fundamental networking device that enhances network performance by dividing a large network into smaller, manageable segments. It plays a critical role in optimizing data transmission within Local Area Networks (LANs).

A bridge is a network device that operates at the Data Link Layer (Layer 2) of the OSI model and is used to connect and segment networks. Its primary function is to filter, forward, or block data frames based on the Media Access Control (MAC) address.

In simple terms, a bridge acts as an intelligent traffic manager, ensuring that data reaches only the intended destination without unnecessary duplication across the network.

Concept of Collision Domain

Before understanding bridging in depth, it is essential to understand the concept of a collision domain.

- A collision domain is a network area where data packets can collide.
- In a large network without segmentation, all devices share a single collision domain, leading to frequent collisions.

A bridge divides the network into multiple collision domains, thereby:

- Reducing data collisions
- Improving efficiency
- Increasing overall throughput

Working Principle of a Bridge

The operation of a bridge is based on the concept of frame filtering and forwarding using MAC addresses.

Step-by-Step Working Process

1. Frame Reception
When a device sends data, it is received by the bridge in the form of a frame.
2. MAC Address Inspection
The bridge reads:
 - Source MAC address
 - Destination MAC address
3. Learning Process
The Bridge stores the source MAC address in a MAC address table (also called a forwarding table).
4. Decision Making
Based on the destination address:
 - If the destination is in the same segment → frame is blocked
 - If in another segment → frame is forwarded
 - If unknown → frame is broadcast
5. Filtering and Forwarding
only necessary data is transmitted to the required segment.

Illustrative Example

Consider a network divided into two segments:

- Segment 1: PC1, PC2
- Segment 2: PC3, PC4

If PC1 sends data to PC2:

- The bridge recognizes both are in the same segment
- The frame is not forwarded to Segment 2

If PC1 sends data to PC3:

- The bridge forwards the frame to Segment 2

This selective forwarding significantly reduces unnecessary traffic.

Functions of a Bridge

A bridge performs several important functions:

1. Network Segmentation
Divides a large network into smaller segments
2. Traffic Filtering
Reduces unnecessary data transmission
3. MAC Address Learning
Maintains a table of device addresses
4. Collision Reduction
Minimizes collisions within segments
5. Bandwidth Optimization
Ensures efficient use of available bandwidth

Types of Bridges

Transparent Bridge

Transparent bridges are widely used in Ethernet networks.

Characteristics:

- Operate invisibly (devices are unaware of their presence)
- Automatically learn MAC addresses
- Use a forwarding table

Working:

- Frames are forwarded based on MAC address
- No configuration required

Translational Bridge

A translational bridge connects different types of network architectures.

Purpose:

- Converts data formats between different protocols

Examples:

- Ethernet to Token Ring

- Copper cable to Fiber optic

Function:

- Adds or removes headers
- Converts frame formats

Source-Route Bridge

This type is used in Token Ring networks, developed by IBM.

Key Feature:

- The entire route is included in the data frame

Working:

- The sender defines the path
- The bridge simply follows the route

Advantages of Bridges

Bridges offer several benefits:

- Reduces network congestion by filtering traffic
- Improves performance through segmentation
- Decreases collisions in the network
- Extends network coverage like a repeater
- Enhances bandwidth utilization
- Connects similar network segments efficiently

Disadvantages of Bridges

Despite their usefulness, bridges have certain limitations:

- Cannot interpret IP addresses (only MAC addresses)
- Cannot effectively connect complex or heterogeneous networks
- Forward all broadcast messages, causing potential congestion
- More expensive than basic devices like repeaters
- Not suitable for large-scale WAN environments

Bridge vs Modern Devices

From a modern networking perspective, bridges have largely been replaced by switches.

Reason:

- Switches perform the same function but with higher speed and efficiency
- Switches support full-duplex communication
- Better handling of large networks

3. Switching

As computer networks evolved, the limitations of shared communication mediums—such as excessive collisions, inefficient bandwidth utilization, and broadcast overhead—became increasingly evident. To overcome these challenges, switching technology emerged as a more advanced and efficient alternative to earlier devices like hubs and bridges.

Switching is now a core mechanism in modern Local Area Networks (LANs), enabling high-speed, reliable, and intelligent data transmission between devices.

Switching is a technique used in computer networks to transfer data from a source device to a destination device efficiently by selecting an appropriate path.

A network switch is a device that operates primarily at the Data Link Layer (Layer 2) of the OSI model and uses MAC addresses to forward data frames. A switch acts as an intelligent multi-port bridge, ensuring that data is sent only to the intended recipient rather than broadcasting it to all devices.

Collision Domain and Broadcast Domain

Collision Domain

- A collision domain is a part of a network where data collisions can occur.
- In switching:
 - Each port of a switch represents a separate collision domain
 - This eliminates collisions almost completely

Broadcast Domain

- A broadcast domain includes all devices that receive broadcast messages.
- Switches do not break broadcast domains (unless VLANs are used)

Thus, switching improves efficiency by reducing collisions while maintaining communication scope.

4. Working Principle of a Switch

The operation of a switch is based on frame forwarding using MAC address tables.

Step-by-Step Working Process

1. Frame Reception
The switch receives a data frame from a device.
2. MAC Address Learning
 - The switch reads the source MAC address
 - Stores it in a MAC address table along with the port number
3. Destination Address checking
the switch checks whether the destination MAC address exists in its table.
4. Forwarding Decision
 - If known → frame is sent only to that specific port
 - If unknown → frame is broadcast to all ports (flooding)
 - If destination is in same port → frame is filtered
5. Data Transmission
Data reaches only the intended device.

Illustrative Example

Consider a switch connecting four devices:

- PC1 (Port 1)
- PC2 (Port 2)
- PC3 (Port 3)
- PC4 (Port 4)

If PC1 sends data to PC3:

- Switch checks MAC table
- Finds PC3 on Port 3
- Sends data only to Port 3

Other devices do not receive the data → efficient communication

5. Functions of a Switch

A switch performs several critical networking functions:

1. Frame Forwarding
Sends data to the correct destination
2. MAC Address Learning
Builds and maintains a MAC address table
3. Filtering
Prevents unnecessary data transmission
4. Collision Reduction
Eliminates collisions by isolating ports
5. Full-Duplex Communication
Allows simultaneous sending and receiving
6. Bandwidth Optimization
Provides dedicated bandwidth per port

Types of Switching Techniques

Circuit Switching

Definition

Circuit switching is a connection-oriented communication method where a dedicated path (circuit) is established between sender and receiver before data transmission begins.

Phases of Circuit Switching

1. Circuit Establishment – Path is created
2. Data Transfer – Continuous data transmission
3. Circuit Disconnect – Path is released

Example

- Traditional telephone systems

Characteristics

- Dedicated path
- Continuous connection
- Reliable communication

Limitation

- Inefficient use of bandwidth (path remains reserved even when idle)

Packet Switching

Definition

Packet switching is a method where data is divided into smaller units called packets, which are transmitted independently across the network.

Key Features

- No dedicated path required
- Packets travel through different routes
- Reassembled at destination

Example

- Internet communication

Advantages of Packet Switching

- Efficient bandwidth utilization
- Fault tolerance (alternative paths available)
- High data transmission quality

Types of Packet Switching

Connectionless Packet Switching (Datagram Switching)

- Each packet is treated independently
- Packets may travel through different paths
- Data may arrive out of order

Reassembly is required at the destination

Connection-Oriented Packet Switching (Virtual Circuit)

- A predefined path is established
- Packets are delivered in sequential order

More reliable than connectionless method

Message Switching

Definition

Message switching is a method where the entire message is sent as a single unit, without establishing a dedicated path.

Working

- Message is sent from sender to intermediate node
- Intermediate node:
 - Stores the message
 - Checks for errors
 - Forwards it to next node

This process continues until the message reaches the destination

Key Feature

- Known as Store-and-Forward Technique

Characteristics

- No dedicated connection required
- Flexible routing

Limitation

- High delay due to storage at each node

Technique	Path Type	Speed	Efficiency	Usage
Circuit Switching	Dedicated	Fast (after setup)	Low	Telephone
Packet Switching	Dynamic	High	High	Internet
Message Switching	No fixed path	Slow	Moderate	Obsolete

Types of Network Switches

Unmanaged Switch

- Simple plug-and-play device
- No configuration required

Managed Switch

- Advanced configuration options
- Supports VLANs, security, monitoring

Layer 3 Switch

- Works at both Layer 2 and Layer 3
- Can perform routing functions

Advantages of Switching

- Eliminates network collisions
- Improves network performance
- Provides dedicated bandwidth
- Supports full-duplex communication
- Enhances security (data sent to specific device)
- Scalable and efficient for large networks

Disadvantages of Switching

- More expensive than hubs
- Configuration complexity in managed switches
- Broadcast traffic can still exist
- Requires proper network design for efficiency

DATA ENCODING

In computer networks, communication between devices requires the transmission of data across various physical media such as copper cables, optical fibres, or wireless channels. However, the data generated by computers is inherently digital, represented in the form of binary digits (0s and 1s). These binary values cannot be transmitted directly in their raw form; instead, they must be converted into signals that can propagate through a transmission medium.

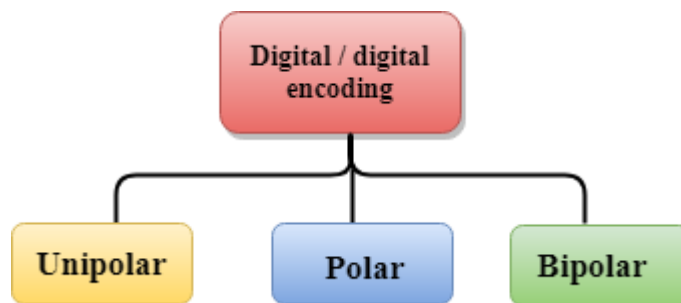
This transformation process is known as data encoding, and it plays a crucial role in ensuring that information is transmitted accurately, efficiently, and reliably from the sender to the receiver. Without encoding, communication systems would fail to maintain synchronization, resist noise, or utilize bandwidth effectively.

Data encoding can be defined as the process of converting data into a signal format suitable for transmission over a communication medium. The encoding process determines how binary data is mapped onto electrical, optical, or electromagnetic signals.

In a typical communication system:

- The sender encodes data into signals
- The channel carries the encoded signals
- The receiver decodes the signals back into original data

Thus, encoding serves as a bridge between abstract data and physical transmission.



3. Importance of Data Encoding

The necessity of data encoding arises from multiple technical challenges associated with data transmission:

3.1 Signal Compatibility

Different transmission media require different types of signals. For example, fiber optics require light signals, while copper cables use electrical signals.

3.2 Synchronization

Proper encoding ensures that the sender and receiver remain synchronized, allowing accurate interpretation of bit boundaries.

3.3 Error Detection and Noise Handling

Encoding techniques help detect and sometimes correct errors caused by noise and interference in the communication channel.

3.4 Efficient Bandwidth Utilization

Efficient encoding reduces the amount of bandwidth required, enabling faster data transmission.

3.5 Reliability of Communication

Encoding enhances the reliability and integrity of transmitted data.

4. Classification of Data Encoding Techniques

Data encoding techniques are broadly classified into four categories based on the nature of data and signals:

4.1 Digital Data to Digital Signal Encoding (Line Coding)

This method involves converting binary data into digital signals suitable for transmission over digital media.

4.1.1 Unipolar Encoding

Unipolar encoding uses a single voltage level to represent binary values.

- Binary 1 → Positive voltage
- Binary 0 → Zero voltage

Although simple, this method suffers from poor synchronization and the presence of a DC component, making it inefficient for long-distance communication.

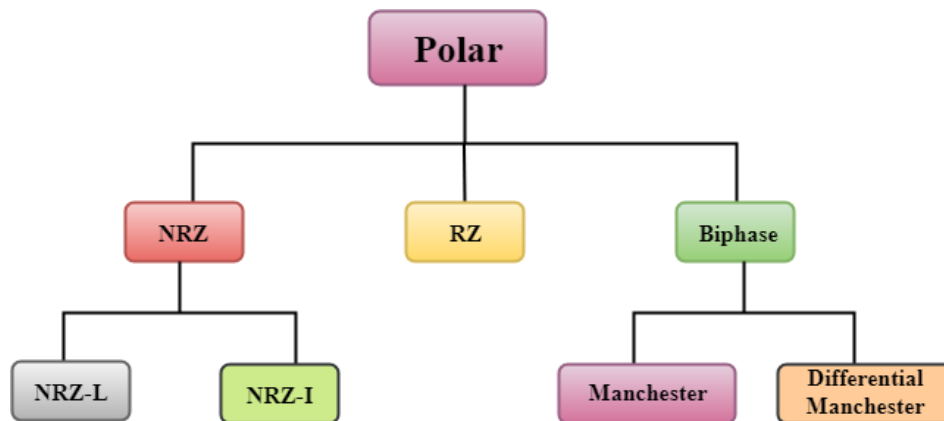
4.1.2 Polar Encoding

Polar encoding uses two voltage levels (positive and negative) to represent binary digits.

Examples include:

- NRZ (Non-Return to Zero): Signal level remains constant during the bit interval
- RZ (Return to Zero): Signal returns to zero between bits

While polar encoding improves signal representation compared to unipolar encoding, it may still encounter synchronization issues in long sequences of identical bits.



In data communication, line coding is a crucial process in which digital data (binary bits) is converted into digital signals for transmission over a communication medium. Among the various encoding techniques, Non-Return to Zero (NRZ), Return to Zero (RZ), and Biphase encoding are widely studied due to their role in synchronization and signal representation.

Each technique differs in how it represents binary values (0 and 1) using voltage levels and signal transitions.

NRZ (Non-Return to Zero)

NRZ (Non-Return to Zero) is a line coding technique in which the signal level remains constant during the entire bit interval and does not return to zero between bits.

- Signal has only two voltage levels:
 - Positive voltage
 - Negative voltage
- No transition to zero between bits

This makes NRZ simple but may create synchronization problems.

Types of NRZ Encoding

NRZ-L (Non-Return to Zero-Level)

Working

In NRZ-L, the voltage level directly represents the bit value:

- Binary 1 → Negative voltage
- Binary 0 → Positive voltage

The signal level depends on the state of the bit.

Characteristics

- Simple to implement
- No transition required between bits

Limitation

- Long sequence of same bits → No transitions → Synchronization loss

NRZ-I (Non-Return to Zero-Invert)

Working

In NRZ-I, the presence or absence of transition determines the bit value:

- Binary 1 → Change in voltage (transition)
- Binary 0 → No change in voltage

Information is encoded in signal transitions, not levels.

Characteristics

- Better synchronization than NRZ-L
- Handles long sequences of 1s effectively

Limitation

- Still problematic for long sequences of 0s

RZ (Return to Zero)

Definition

RZ (Return to Zero) is an encoding technique in which the signal returns to zero level in the middle of each bit interval.

Working

- Binary 1 → Positive voltage → Returns to zero
- Binary 0 → Negative voltage → Returns to zero

Each bit has two signal changes:

1. From voltage to zero

2. Start of next bit

Key Characteristics

- Uses three levels:
 - Positive
 - Negative
 - Zero
- Ensures better synchronization

Advantage

- Easy clock recovery due to frequent transitions

Disadvantage

- Requires higher bandwidth
- Inefficient due to double transitions per bit

Biphase Encoding

Definition

Biphase encoding is a technique where there is always a transition at the middle of each bit interval, ensuring synchronization.

Unlike RZ, it does not necessarily return to zero, but guarantees a transition.

Types of Biphase Encoding

Manchester Encoding

Working

- Binary 1 → Low to High transition (negative to positive)
- Binary 0 → High to Low transition (positive to negative)

Transition occurs in the middle of each bit

Characteristics

- Self-synchronizing
- No DC component
- Reliable transmission

Advantage

- Excellent synchronization
- Widely used in Ethernet

Disadvantage

- Requires double bandwidth compared to NRZ

Differential Manchester Encoding

Working

- Always has a transition at the middle (for synchronization)
- Bit value determined at the start of interval:
 - Transition → Binary 0
 - No transition → Binary 1

Characteristics

- More resistant to noise
- Independent of voltage polarity

Clarification

- Binary 0 → Two transitions
- Binary 1 → One transition

4.1.3 Bipolar Encoding

Bipolar encoding introduces three voltage levels: positive, negative, and zero.

- Binary 1 → Alternating positive and negative voltages
- Binary 0 → Zero voltage

This technique reduces the DC component and improves synchronization, making it more efficient than previous methods.

4.1.4 Manchester Encoding

Manchester encoding combines data and clock information within the signal.

- Binary 0 → High-to-low transition
- Binary 1 → Low-to-high transition

This ensures self-synchronization and eliminates DC components, although it requires higher bandwidth.

4.1.5 Differential Manchester Encoding

This method relies on transitions rather than voltage levels, making it more robust against noise and signal distortion.

4.2 Digital Data to Analog Signal Encoding (Modulation)

When digital data needs to be transmitted over analog media, it is converted into analog signals using modulation techniques.

4.2.1 Amplitude Shift Keying (ASK)

- Amplitude of the signal varies according to the data
- Simple but highly susceptible to noise

4.2.2 Frequency Shift Keying (FSK)

- Frequency changes to represent binary values
- More resistant to noise compared to ASK

4.2.3 Phase Shift Keying (PSK)

- Phase of the signal changes based on data
- Provides better performance and reliability

PSK and its advanced forms are widely used in modern communication systems due to their efficiency.

4.3 Analog Data to Digital Signal Encoding

This type of encoding is used when analog signals (such as voice) must be converted into digital form for transmission or storage.

4.3.1 Pulse Code Modulation (PCM)

- Analog signal is sampled, quantized, and encoded into binary form
- Widely used in digital telephony

4.3.2 Delta Modulation (DM)

- Encodes the difference between successive samples
- Simpler but less accurate than PCM

4.4 Analog Data to Analog Signal Encoding

This method is used in traditional communication systems where analog signals are transmitted directly.

Techniques include:

- Amplitude Modulation (AM)
- Frequency Modulation (FM)
- Phase Modulation (PM)

These techniques are commonly used in radio and television broadcasting.

5. Key Concepts in Data Encoding

5.1 Bandwidth

Bandwidth refers to the range of frequencies available for transmission. Efficient encoding minimizes bandwidth usage.

5.2 Bit Rate and Baud Rate

- Bit Rate: Number of bits transmitted per second
- Baud Rate: Number of signal changes per second

5.3 Noise and Interference

Encoding techniques are designed to minimize the impact of noise on data transmission.

5.4 Synchronization

Maintaining synchronization between sender and receiver is essential for accurate decoding.

6. Advantages of Data Encoding

- Enhances data transmission reliability
- Reduces errors and interference

- Improves signal quality
- Enables efficient bandwidth usage
- Supports high-speed communication

7. Disadvantages of Data Encoding

- Some techniques require additional bandwidth
- Increased system complexity
- Higher implementation cost

8. Applications of Data Encoding

Data encoding is widely used in:

- Internet and network communication
- Mobile communication systems
- Satellite transmission
- Digital audio and video systems
- Wireless communication technologies

Among the various techniques, Manchester encoding and Phase Shift Keying (PSK) are often preferred due to their robustness and synchronization capabilities. However, the trade-off between bandwidth efficiency and reliability remains a critical design consideration in network engineering.

Data encoding is an essential mechanism that enables the transformation of digital information into transmittable signals. By ensuring synchronization, minimizing errors, and optimizing bandwidth usage, encoding plays a vital role in the successful operation of computer networks. As communication technologies continue to evolve, data encoding techniques will remain central to achieving efficient and reliable data transmission.

Ethernet Switch

In modern computer networks, efficient communication between devices is essential. Among the various networking devices, the Ethernet switch plays a central role in enabling high-speed and reliable data transfer within Local Area Networks (LANs).

Although multiple communication standards exist, Ethernet has become the most widely adopted networking technology, making Ethernet switches a fundamental component of almost every data network. The importance of

switches is evident from the fact that network failures often occur when a switch fails, highlighting their critical role in maintaining connectivity.

What is an Ethernet Switch?

An Ethernet switch is a networking device used to connect multiple devices within the same network and facilitate communication between them.

Connected Devices (Nodes) Include:

- Computers
- Servers
- Printers
- Wireless Access Points
- Other switches

An Ethernet switch acts as an intelligent device that:

- Receives data from one device
- Identifies the destination
- Sends the data only to the intended device

This ensures efficient communication without unnecessary data transmission.

3. Key Characteristics of Ethernet Switch

- Operates mainly at the Data Link Layer (Layer 2)
- Uses MAC (Media Access Control) addresses
- Provides dedicated bandwidth per port
- Reduces network congestion
- Works as a multi-port bridge

4. Types of Ethernet Switches

4.1 Unmanaged Switch

An unmanaged switch is a basic type of Ethernet switch that operates without manual configuration.

Features:

- Plug-and-play device
- No setup required
- Low cost

- Used in small networks (home, labs, offices)

Suitable for simple connectivity needs.

4.2 Managed Switch

A managed switch offers advanced features and control over network operations.

Features:

- Can be configured by administrators
- Supports security mechanisms
- Enables traffic prioritization (QoS)
- Improves performance and monitoring

Suitable for large and complex networks.

4.3 Industrial Switch

Industrial switches are designed for harsh environments.

Features:

- High durability
- Operates in extreme temperatures
- High reliability (carrier-grade performance)

Used in industries and production environments.

5. Working of an Ethernet Switch

Ethernet switches work by forwarding data frames between connected devices.

Step-by-Step Working

1. Frame Reception

The switch receives an Ethernet frame from a device.

2. MAC Address Reading

It reads:

- Source MAC address
- Destination MAC address

3. Address Learning

The switch stores the source MAC address in a MAC table along with the port number.

4. Forwarding Decision

- If destination is known → send to specific port
- If unknown → send to all ports (flooding)
- If same port → filter the frame

5. Data Transmission

The frame is delivered to the correct device.

6. Ethernet Frames

Communication in Ethernet networks takes place using Ethernet frames.

Structure of an Ethernet Frame

1. Header

Contains:

- Preamble (synchronization)
- Destination MAC address
- Source MAC address
- Type/Length field

2. Payload

- Actual data being transmitted

3. Trailer

- Frame Check Sequence (FCS) for error detection

This structured format ensures reliable communication.

7. Ethernet Switching Concept (Bridging)

Ethernet switches use a concept called transparent bridging, standardized by IEEE 802.1D.

Key Idea

- Frames pass through the switch without modification
- Devices are unaware of the switch's presence

This makes switching transparent and automatic.

8. MAC Address Learning Mechanism

One of the most important features of an Ethernet switch is address learning.

How it Works:

- Each device has a unique 48-bit MAC address
- Switch observes incoming frames
- Stores source addresses in a table

Over time, the switch builds a database of device locations

Promiscuous Mode

Unlike normal devices, a switch:

- Accepts all frames
- Processes frames meant for other devices

This allows efficient forwarding decisions.

9. Traffic Management in Ethernet Switch

Ethernet switches manage network traffic using several techniques:

9.1 Traffic Filtering and Forwarding

- Sends frames only to required ports
- Reduces unnecessary traffic

9.2 Frame Buffering

- Stores frames temporarily if:
 - Destination device is busy
 - Network is congested

9.3 Frame Flooding

- Occurs when destination MAC is unknown
- Frame is sent to all ports except source

Helps discover new devices in the network

9.4 Adaptive Filtering

- Uses MAC table to optimize forwarding decisions
- Minimizes network load

10. Advantages of Ethernet Switch

- High-speed data transmission
- Reduces collisions
- Efficient bandwidth utilization
- Improves network performance
- Secure communication (data sent only to intended device)
- Scalable for large networks

11. Disadvantages of Ethernet Switch

- More expensive than hubs
- Requires configuration (managed switches)
- Broadcast traffic may still exist
- Network design complexity in large systems